Course Code	CS325
Course Title	**Data Science**
CrHr (TCH + LCH)	3(3+0)
Pre-requisite	None
Recommended Texts	1. Data Science from Scratch: First Principles with Python, Joel Grus, O'Reilly Media; 2nd edition (May 16, 2019), ISBN-13:978-1492041139 2. What Is Data Science, Mike Loukides, O'Reilly Media publisher, 2016, ISBN-13: B007R8BHAK 3. Cathy O'Neil and Rachel Schutt. Doing Data Science, Straight Talk from the Frontline. O'Reilly. 2014, ISBN-978-1449358655. 4. Jure Leskovek, Anand Rajaraman and Jeffrey Ullman. Mining of Massive Datasets. v2.1, Cambridge University Press. 2014, ISBN-1107015359.
Course Description	Data Science is the study of the generalizable extraction of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases and other branches of computer science along with a good understanding of the craft of problem formulation to engineer effective solutions. This course will introduce students to this rapidly growing field and equip them with some of its basic principles and tools as well as its general mindset. Students will learn concepts, techniques and tools they need to deal with various facets of data science practice, including data collection and integration, exploratory data analysis, predictive modeling, descriptive modeling, data product creation, evaluation, and effective communication. The focus in the treatment of these topics will be on breadth, rather than depth, and emphasis will be placed on integration and synthesis of concepts and their application to solving problems. To make the learning contextual, real datasets from a variety of disciplines will be used.
Course Objectives	• Describe what Data Science is and the skill sets needed to be a data scientist. • Explain in basic terms what Statistical Inference means. Identify probability distributions commonly used as foundations for statistical modeling. Fit a model to data. • Use R to carry out basic statistical modeling and analysis. • Explain the significance of exploratory data analysis (EDA) in data science. Apply basic tools (plots, graphs, summary statistics) to carry out EDA. • Describe the Data Science Process and how its components interact. • Use APIs and other tools to scrap the Web and collect data.
Course Outline	• Introduction

	• Statistical Interface • Exploratory Data Analysis and the Data Science Process • Data Visualization • Data Science and Ethical Issues

Week Wise Distribution of the Contents

Week Number	Topic
W1	Introduction: What is Data Science: Big Data and Data Science hype and getting past the hype, - Why now? Data fication, Current landscape of perspectives, - Skill sets needed
W2	Statistical Inference,
W3	Populations and samples, Statistical modeling, probability distributions
W4	fitting a model, Intro to R
W5	Exploratory Data Analysis and the Data Science Process, Basic tools (plots, graphs and summary statistics) of EDA , Philosophy of EDA, The Data Science Process, Case Study: RealDirect (online real estate firm) .
W6	Three Basic Machine Learning Algorithms, Linear Regression, k-Nearest Neighbors (k-NN) , k-means
W7	One More Machine Learning Algorithm and Usage in Applications: Motivating application: Filtering Spam, Why Linear Regression and k-NN are poor choices for Filtering Spam.
W8	Naive Bayes and why it works for Filtering Spam, Data Wrangling: APIs and other tools for scrapping the Web
MID EXAM / MID OF SEMESTER	
W9	Feature Generation and Feature Selection (Extracting Meaning from Data): - Motivating application: user (customer) retention,
W10	Feature Generation (brainstorming, role of domain expertise, and place for imagination), Feature Selection algorithms, Filters; Wrappers; Decision Trees; Random Forests
W11	Recommendation Systems: Building a User-Facing Data Product: Algorithmic ingredients of a Recommendation Engine.
W12	Dimensionality Reduction, Singular Value Decomposition, Principal Component Analysis, Exercise: build your own recommendation system.
W13	Mining Social Network Graphs: Social networks as graphs, Clustering of graphs
W14	Direct discovery of communities in graphs, partitioning of graphs, Neighborhood properties in graphs, Direct discovery of communities in graphs, Partitioning of graphs, Neighborhood properties in graphs.
W15	Data Visualization: Basic principles, ideas and tools for data visualization, Examples of inspiring (industry) projects, Exercise: create your own visualization of a complex dataset.
W16	Data Science and Ethical Issues, Discussions on privacy, security, ethics, A look back at Data Science, Next-generation data scientists